TSA Final Project - Forecasting Post-War Electricity Mix in Germany

Github

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Introduction

Several European countries, including Germany, had relied heavily on imported fossil fuels, particularly natural gas from Russia, before the Ukraine-Russia war broke out in February 2023 (Maliszewska-Nienartowicz, 2024). According to data on German gas imports in 2020 (BP), 55.2% of imported gas came from Russia (Umbach, 2022). However, the onset of the war disrupted the energy market across Europe, especially impacting Germany's energy supply. In response, Germany reactivated coal plants, shifting back from natural gas—considered a cleaner energy source—to coal (Luschini, Rego & Montebello, 2024). Significant changes in Germany's energy mix and market were anticipated, and the energy crisis highlighted the urgent need for a transition to renewable energy to enhance energy security by reducing dependence on imported energy sources (Maliszewska-Nienartowicz, 2024).

Under Germany's Renewable Energy Act, the country's renewable energy share of the total electricity generation has reached over 62% in 2024, hitting the historical highest (Collins et al., 2025). It is speculated that the Russia-Ukraine war has accelerated Germany's renewable energy transition (Curry, 2022). In this context, we aimed to investigate how the Ukraine-Russia war (2023) actually affected Germany's energy mix, particularly regarding Natural Gas and Renewables.

Motivations & Relevance

This project focuses specifically on changes in Germany's electricity generation mix. Although Natural Gas plays a primary role in heating and cooling—accounting for 44% of total imported natural gas usage—this study sought to forecast how the role of Natural Gas may evolve in Germany's electricity sector and the subsequent impacts on the adoption of Renewables post Ukraine-Russia war. Forecasting how the Natural Gas and Renewable Energy generation will change in the subsequent year (2025) will help policymakers make decisions regarding the country's energy security and clean energy transition goals.

Objectives

1. Understand how the electricity generation of Renewables and Natural Gas has changed over time;
2. Interpret Germany's electricity mix and how it might be affected by the Russia-Ukraine War;
3. Estimate how the electricity generation of Renewables and Natural Gas will change in 2025.

Dataset information

The entire analysis is based on the Monthly Electricity Production data from the International Energy Agency (IEA)'s Monthly Electricity Statistics in csv format. The dataset consists of the following columns: Country, Time, Balance, Product, Value, and Unit. It contains monthly electricity mix data by energy source from January 2010 to December 2024, covering countries such as Australia, Denmark, Germany, Iceland, Ireland, Japan, and Korea, among others. In the "Product" column, energy sources include Hydro, Coal/Peat & Manufactured Gas, Oil and Petroleum Products, Natural Gas, and Total Renewables (comprising Hydro, Geothermal, Solar, Wind, and Other). Our team cleaned the dataset and filtered it to include only Germany, Total Renewables, and Natural Gas. All units are in GWh.

Table 1: Summary Table for Energy Mix in Germany

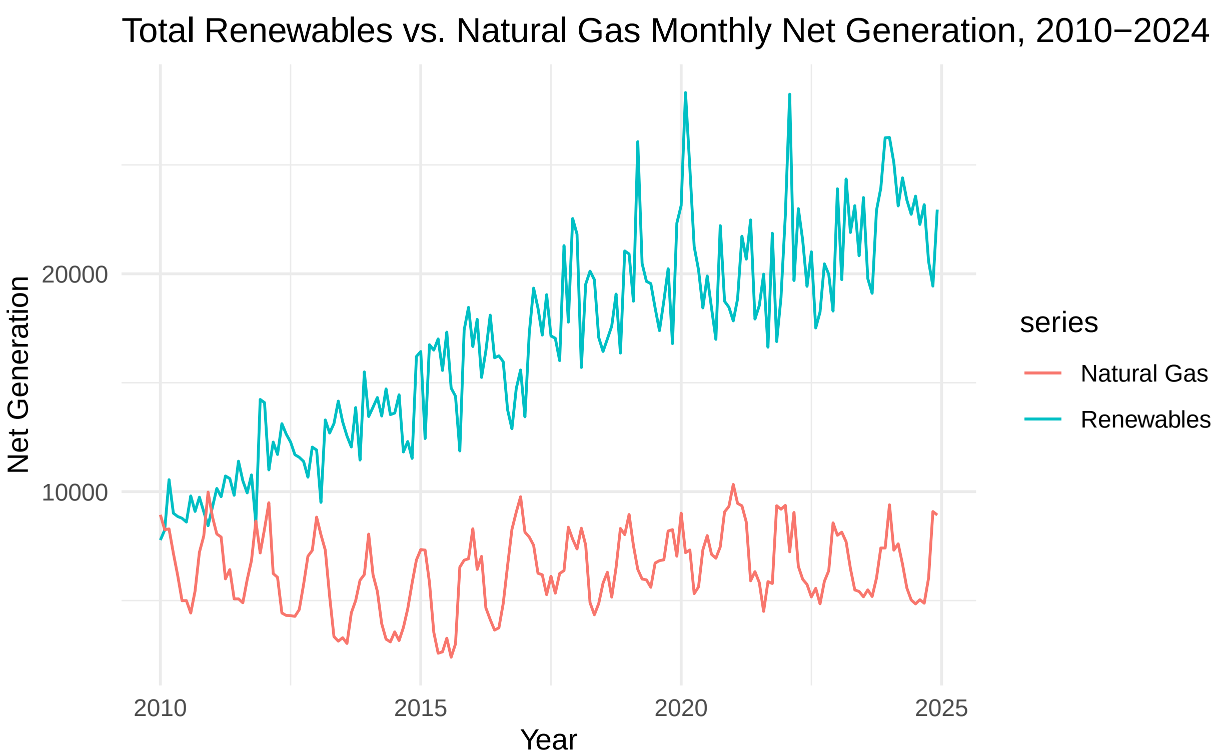
Variable	Natural_Gas	Renewables
Range	2010/01/01 - 2024/12/31	2010/01/01 - 2024/12/31
Unit	GWh	GWh
Central Tendency	Mean: 6419.49	Mean: 16835.20
Source	IEA	IEA

Methods

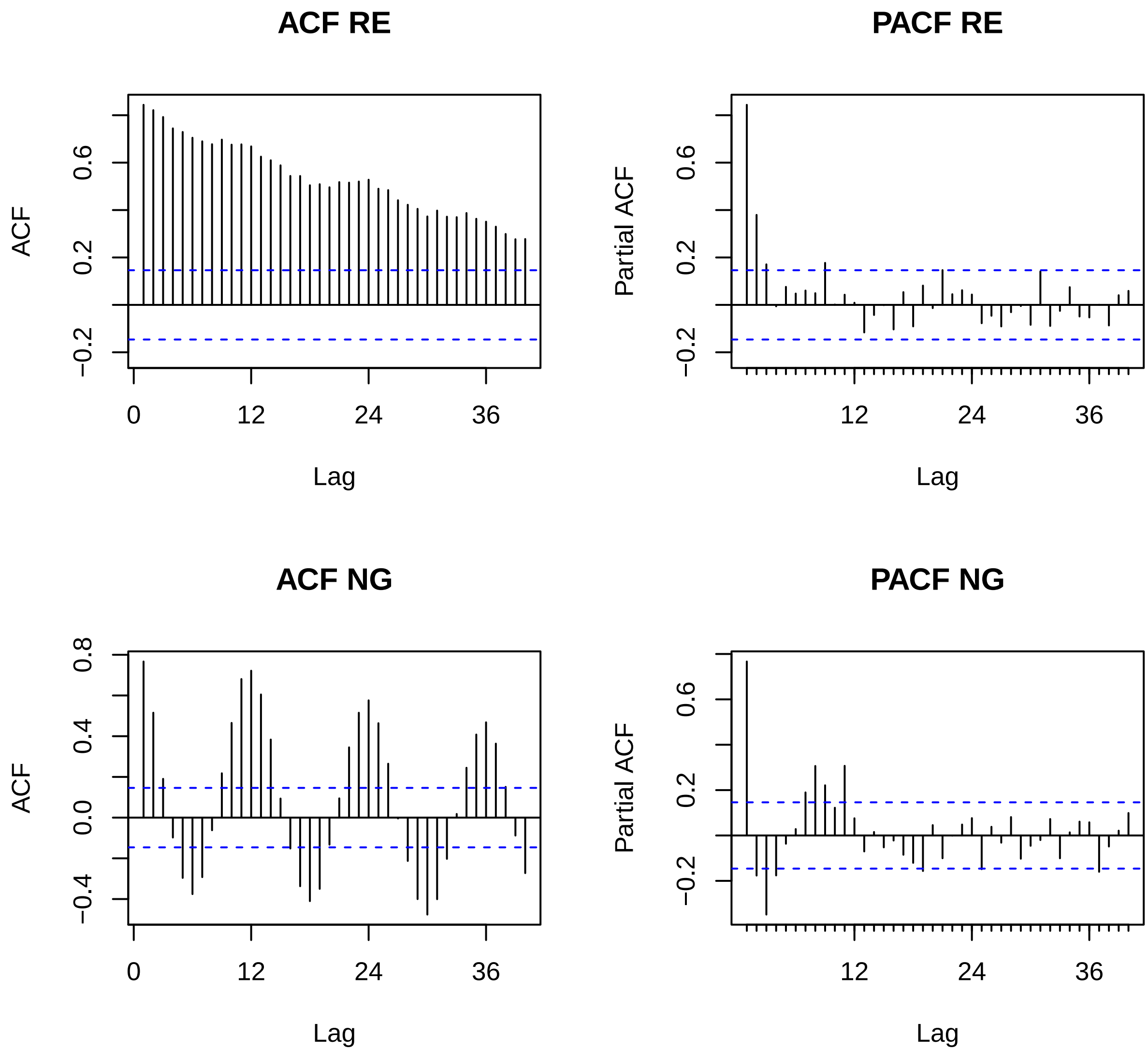
To forecast electricity generation from Natural Gas (NG) and Renewables (RE), we employed a structured time series analysis approach combining exploratory data analysis (EDA), autocorrelation diagnostics, multiple forecasting models, and model accuracy evaluation.

Exploratroy Data Analysis

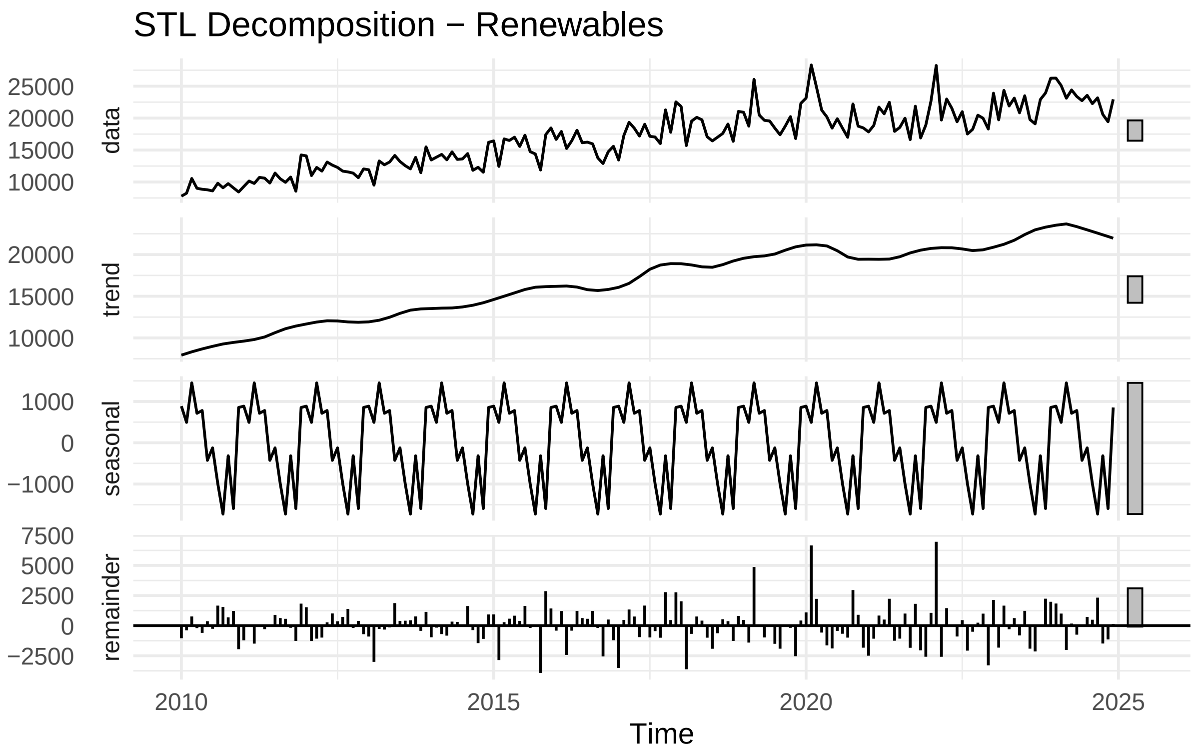
Our EDA began with visual inspections of the **raw generation trends**, which revealed a clear upward trajectory in renewables and pronounced seasonal fluctuations in natural gas.



We then conducted **ACF and PACF analyses** to investigate the underlying temporal structure. For RE, the ACF showed slow decay and no sharp seasonal lags, suggesting a strong trend but weak or irregular seasonality. In contrast, NG displayed prominent spikes at lags 12, 24, and 36, indicating strong annual seasonality and cyclical demand patterns.



STL decomposition was used to break down the time series of renewable and natural gas electricity generation into trend, seasonal, and remainder components. For renewables, the decomposition revealed a strong upward trend. However, the seasonal component was relatively modest in magnitude and showed irregular patterns, indicating weak to moderate seasonality. This is further quantified by the seasonal strength score of 0.3, which falls below the commonly used threshold of 0.6 for strong seasonality. In contrast, the STL decomposition for natural gas showed highly regular and dominant seasonal patterns, with clearly defined annual cycles aligned with heating and cooling demand. The trend component fluctuated over time, reflecting shifts in energy market dynamics and possible substitution effects. The seasonal strength score of 0.8 confirms that natural gas generation exhibits strong and consistent seasonality.

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## Estimated Seasonal Strength for Renewables: 0.289
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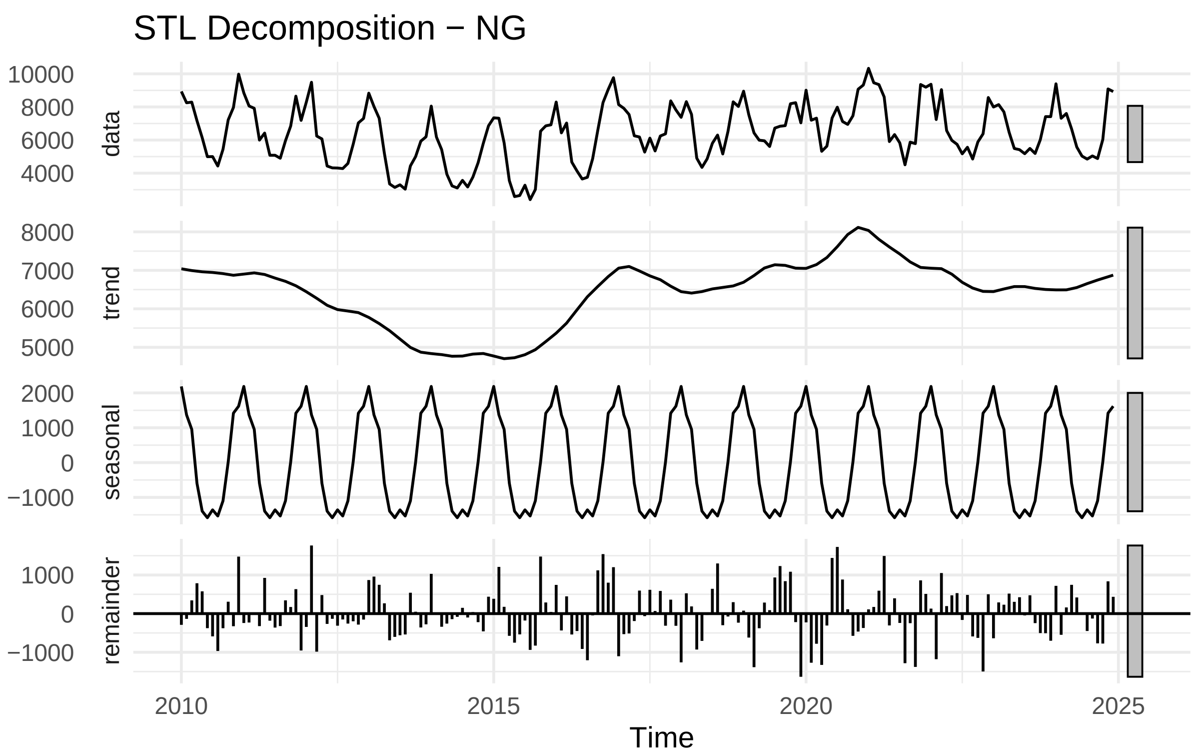


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## Estimated Seasonal Strength for Natural Gas: 0.8
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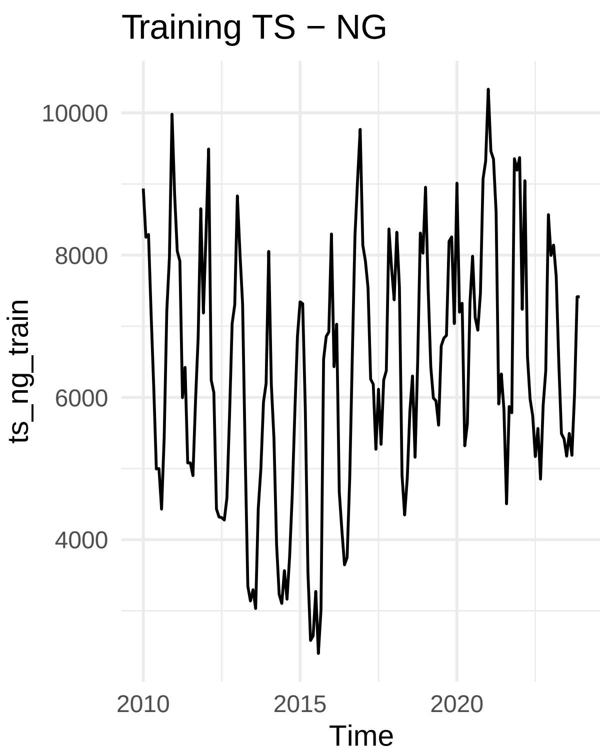
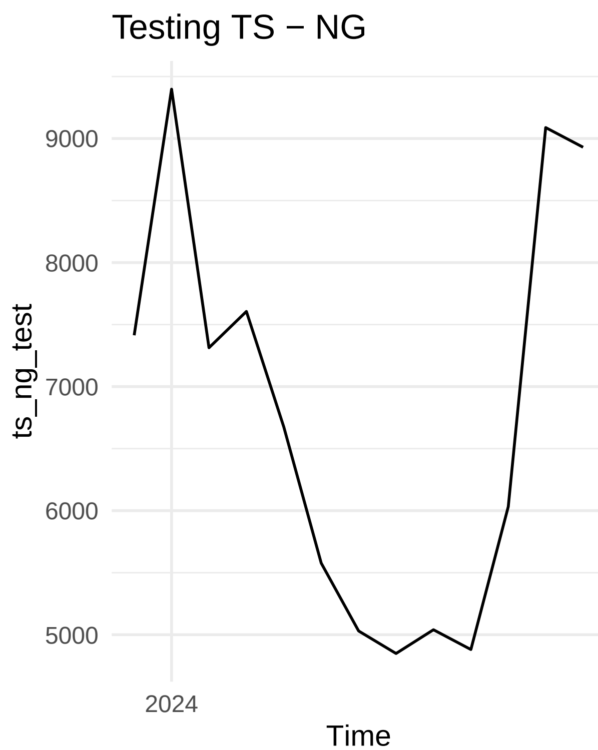
Model Description

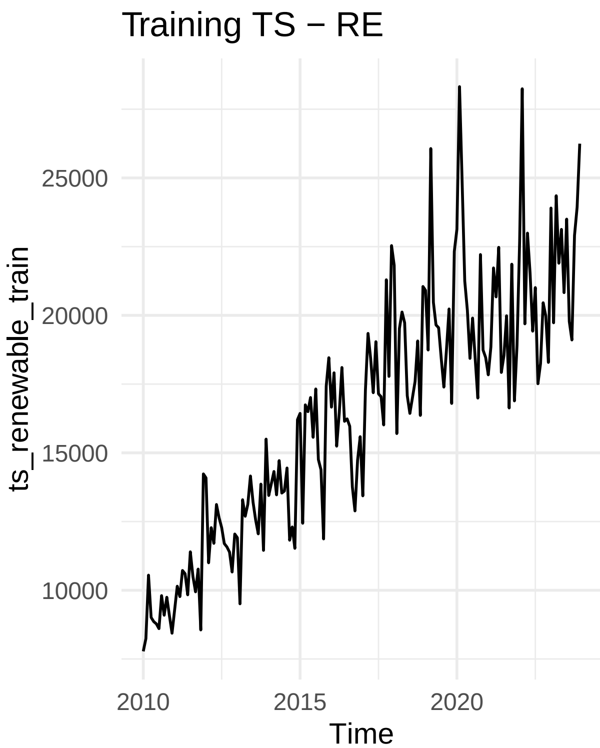
Based on these insights, we selected four forecasting models for testing purpose:

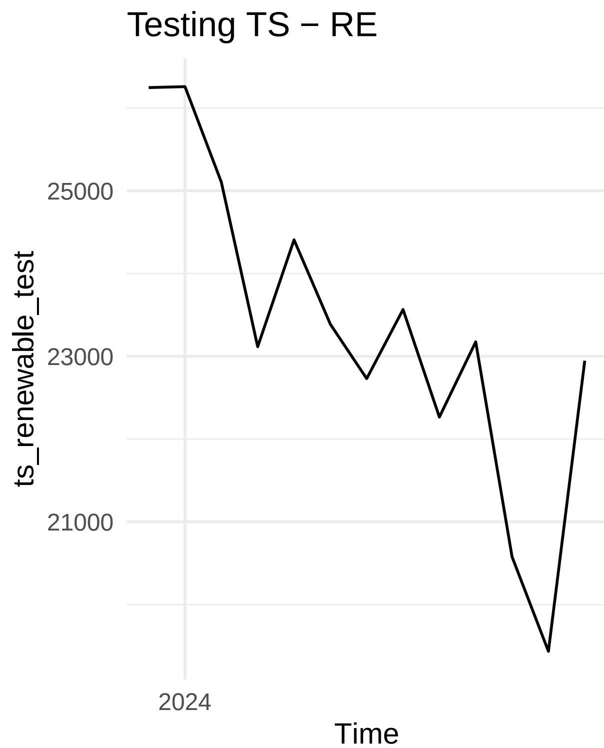
1. **Baseline ARIMA**: We began with a baseline ARIMA model, which captures autoregressive and moving average dependencies in the data.
2. **ARIMA with Fourier terms**: To better accommodate irregular or weak seasonal patterns in renewables, we extended ARIMA with Fourier terms as exogenous regressors, enabling the model to approximate periodic behaviors without enforcing strict seasonality.
3. **STL+ETS**: This model decomposes the time series into trend, seasonal, and remainder components before applying exponential smoothing for forecasting, which is especially useful for data with evolving seasonal patterns.
4. **TBATS**: Trigonometric, Box-Cox, ARMA errors, Trend, and Seasonal. This model supports complex seasonal structures and proved particularly effective for renewable series.
5. **Neural Network Autoregression (NNAR)**: This models nonlinear relationships and includes lagged values and optional Fourier terms to handle multiple seasonality.

The data in time range 2010-2023 is used for training, and 2024 data is used for testing. The model results are then evaluated based on accuracy metrics including RMSE, MAE, MAPE, ACF1, and Theil's U. The best-performing model is selected to forecast monthly generation in 2025.





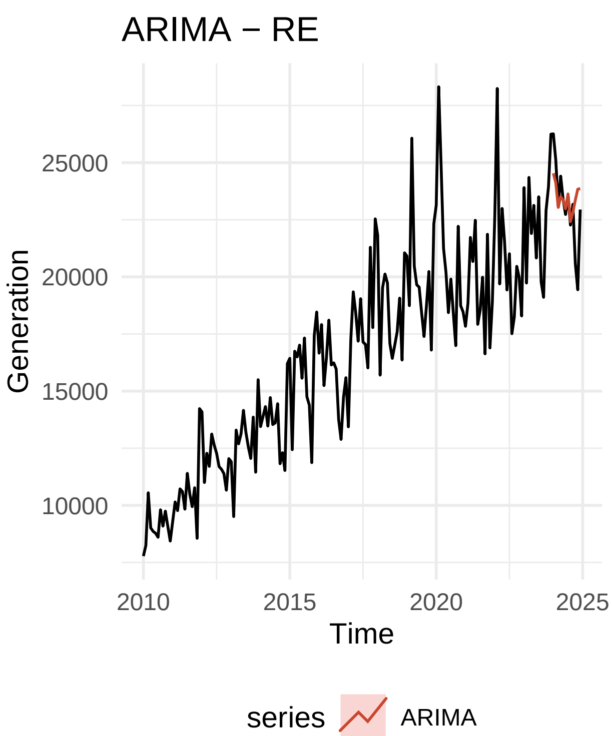


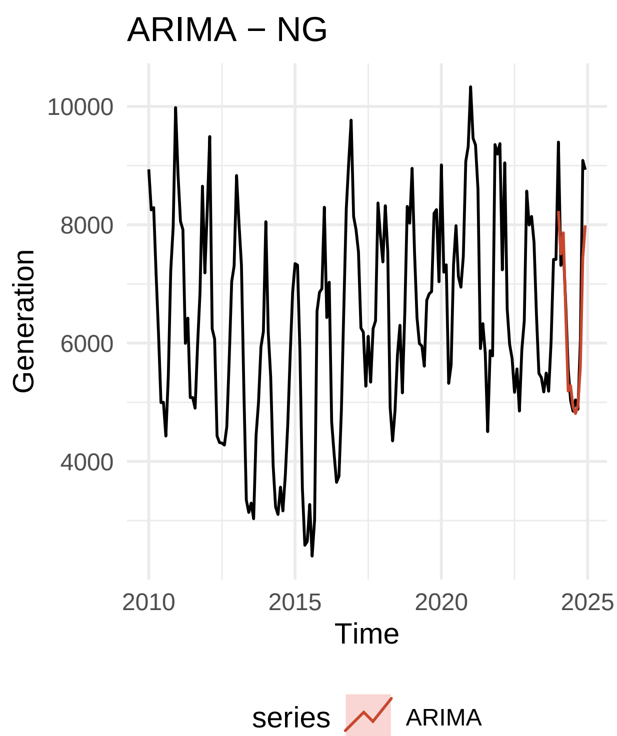


Analysis

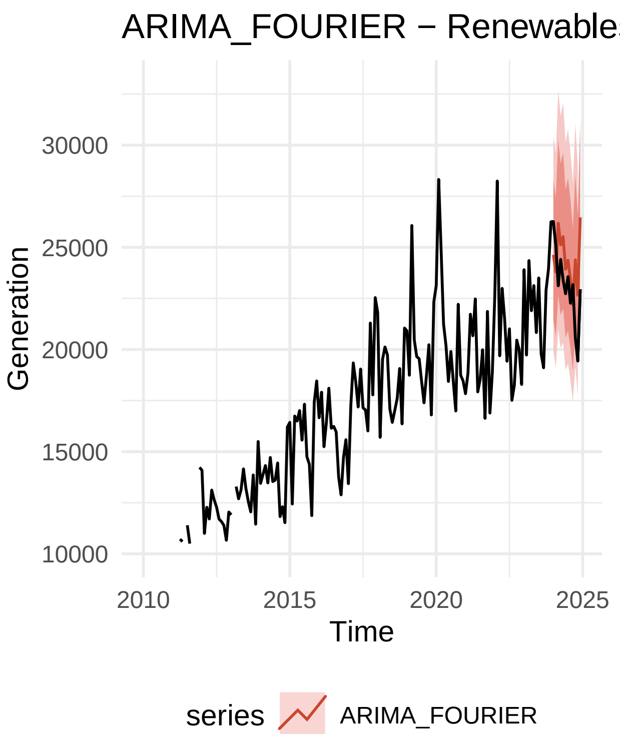
Baseline ARIMA

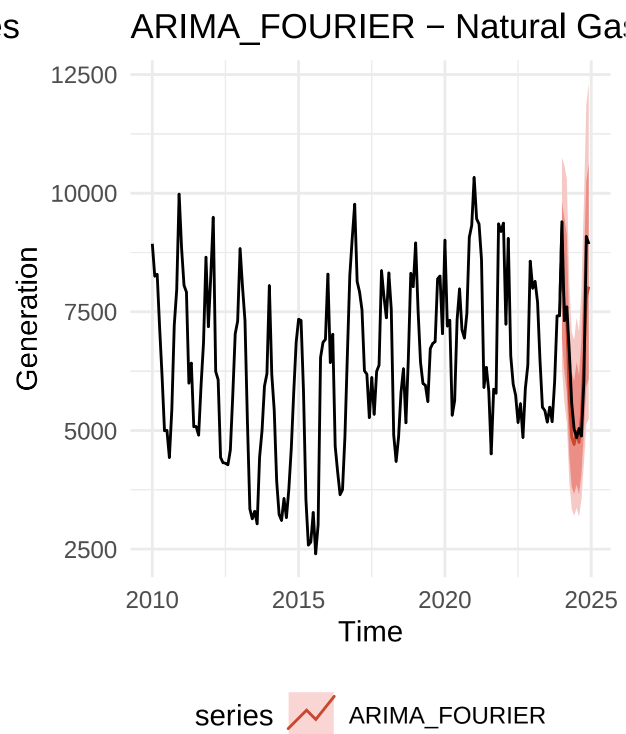
Using auto.arima(), (p,d,q) and (P,D,Q) are (1,1,1) and (0,0,2) for Renewables, and (2,0,2) and (2,1,0) for Natural Gas.



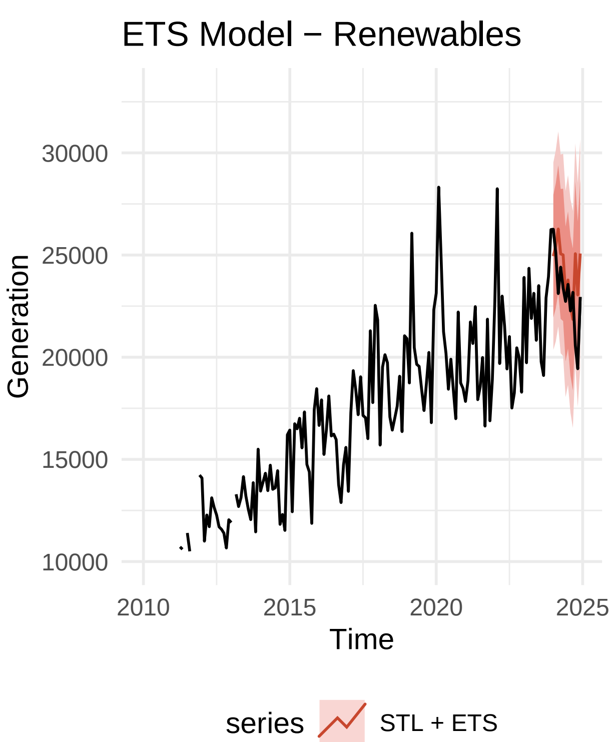


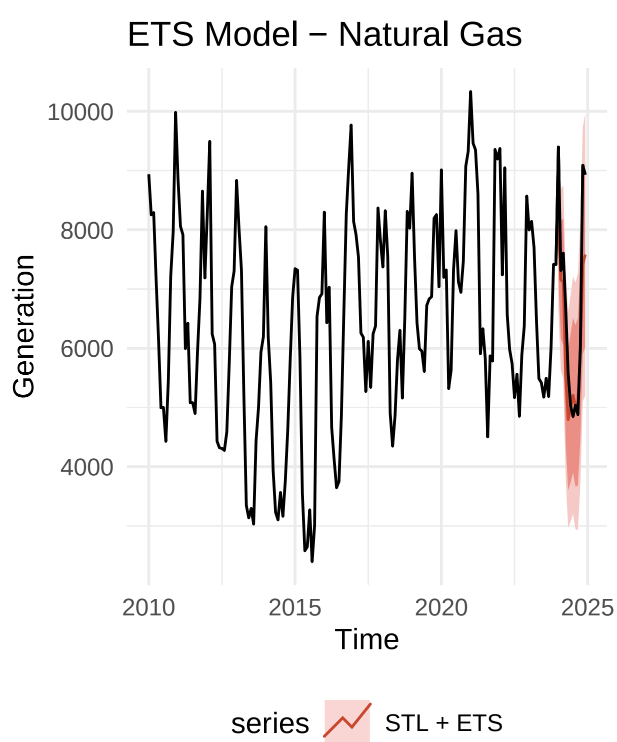
NG - Model 2: ARIMA + FOURIER terms



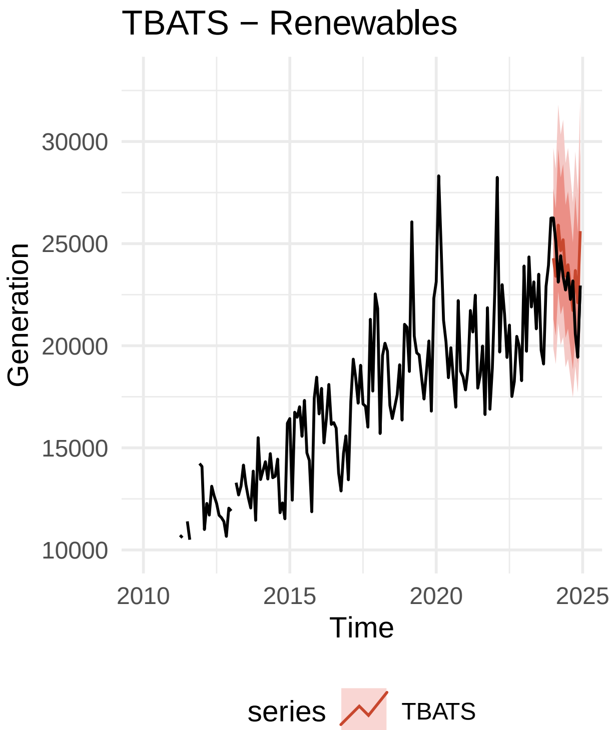


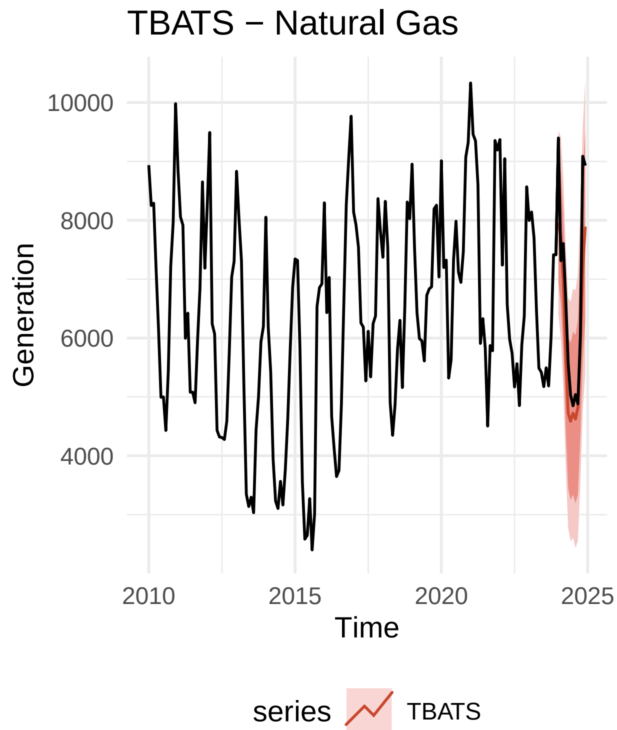
NG - Model 3: STL + ETS



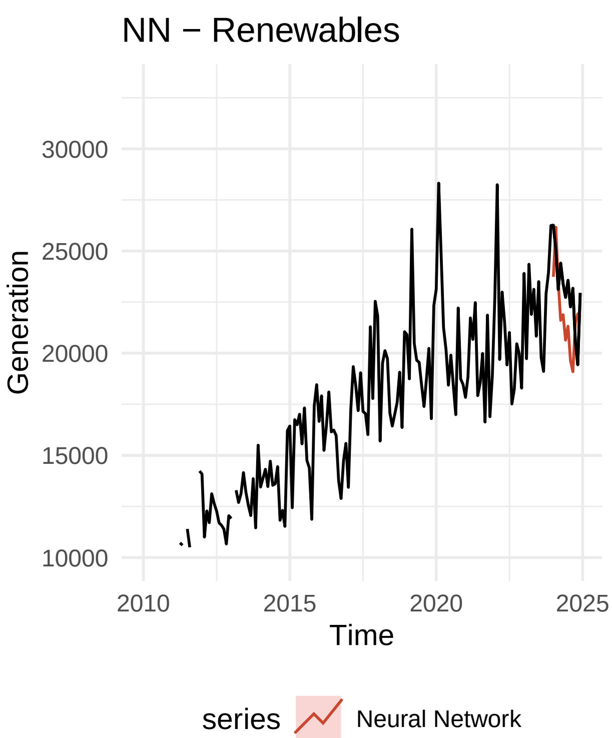


Model 4: TBATS





Model 5: Neural Network Time Series Forecasts



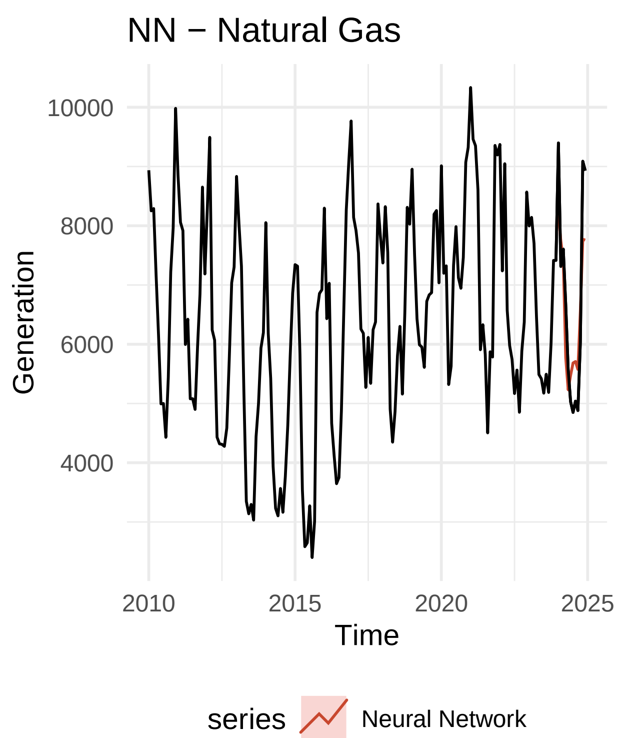


Table 2: Forecast Accuracy for Renewables Monthly Generation

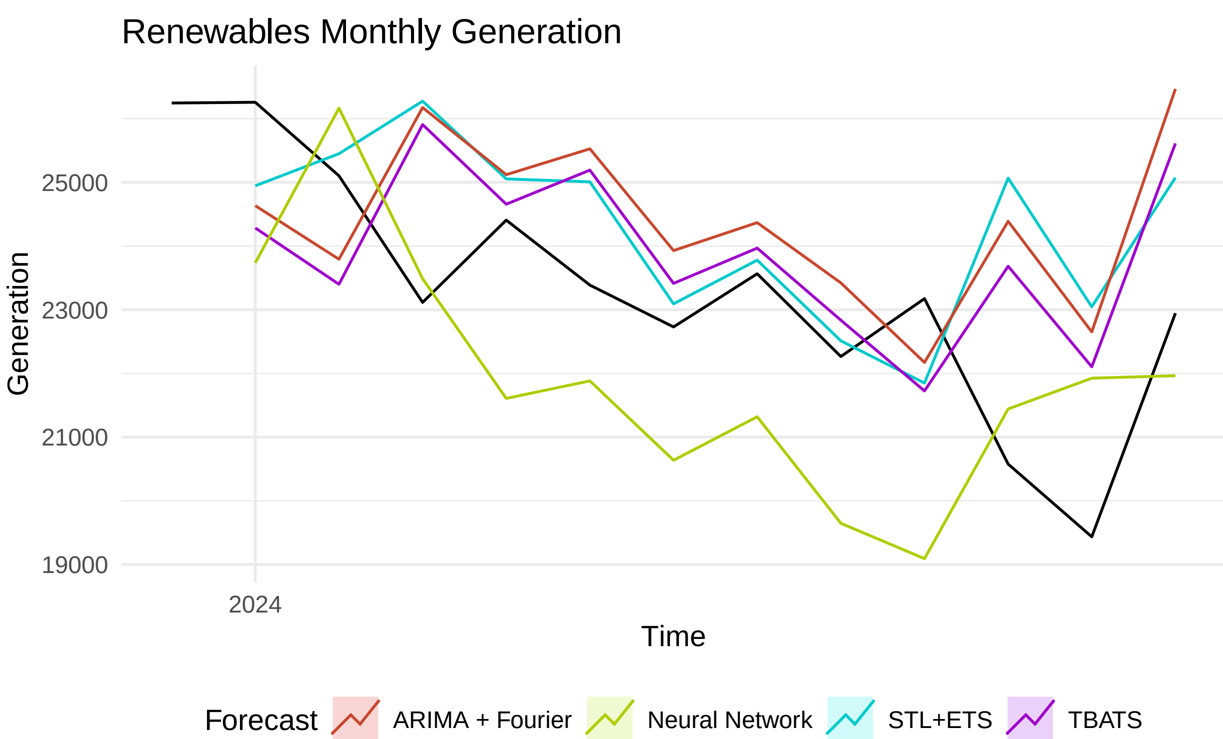
	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
STL + ETS	-1182.3477	2132.696	1621.720	-5.59964	7.38485	0.12871	1.24392
ARIMA + Fourier	-1306.9211	2245.169	1962.840	-6.15580	8.77666	0.12924	1.31884
TBATS	-818.3496	1933.017	1672.475	-3.99966	7.42457	0.09346	1.09499
Neural Network	1171.3434	2209.201	1969.388	4.75303	8.56173	0.18403	1.22685

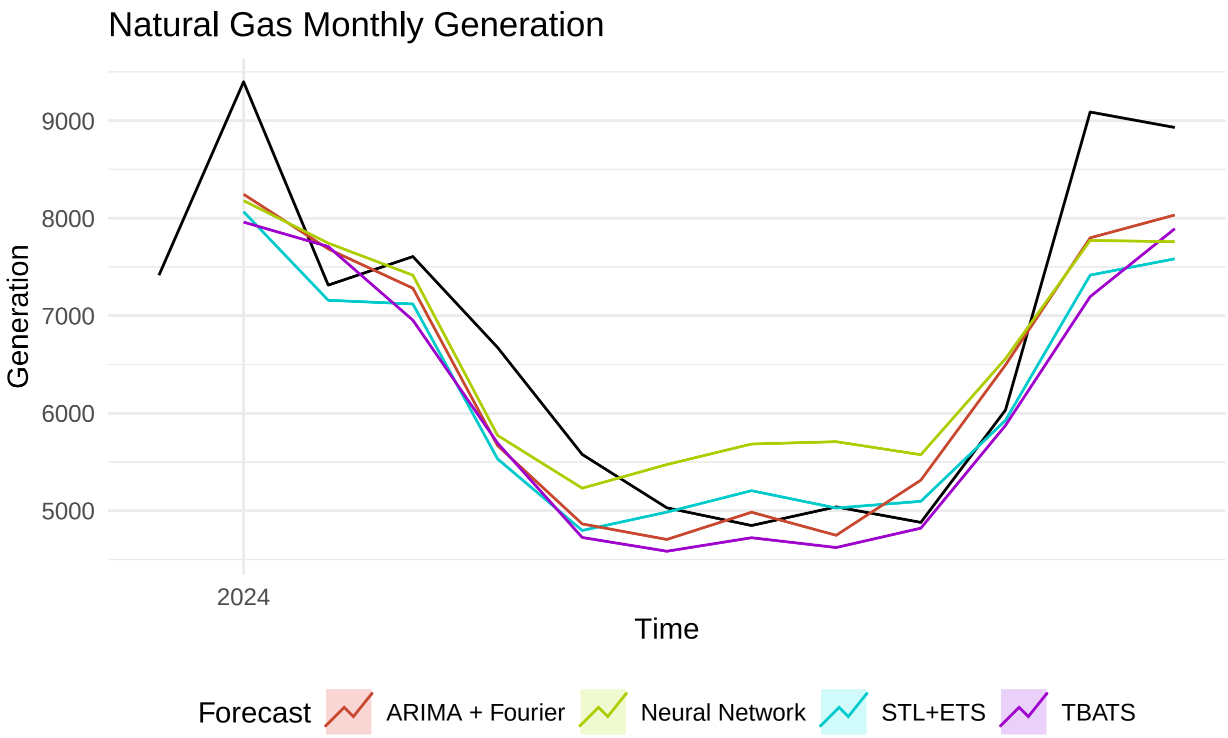
Table 3: Forecast Accuracy for Natural Gas Monthly Generation

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
STL + ETS	541.7984	853.6627	637.3654	6.52604	8.49220	0.34090	0.59263
ARIMA + Fourier	383.0291	717.1232	616.7916	4.70277	8.77286	0.03534	0.51703
TBATS	638.4602	885.7547	704.3757	8.69992	9.60123	-0.06374	0.63652
Neural Network	128.6809	808.5432	728.6810	-0.44929	10.91271	0.31688	0.63200

Accuracy Evaluation

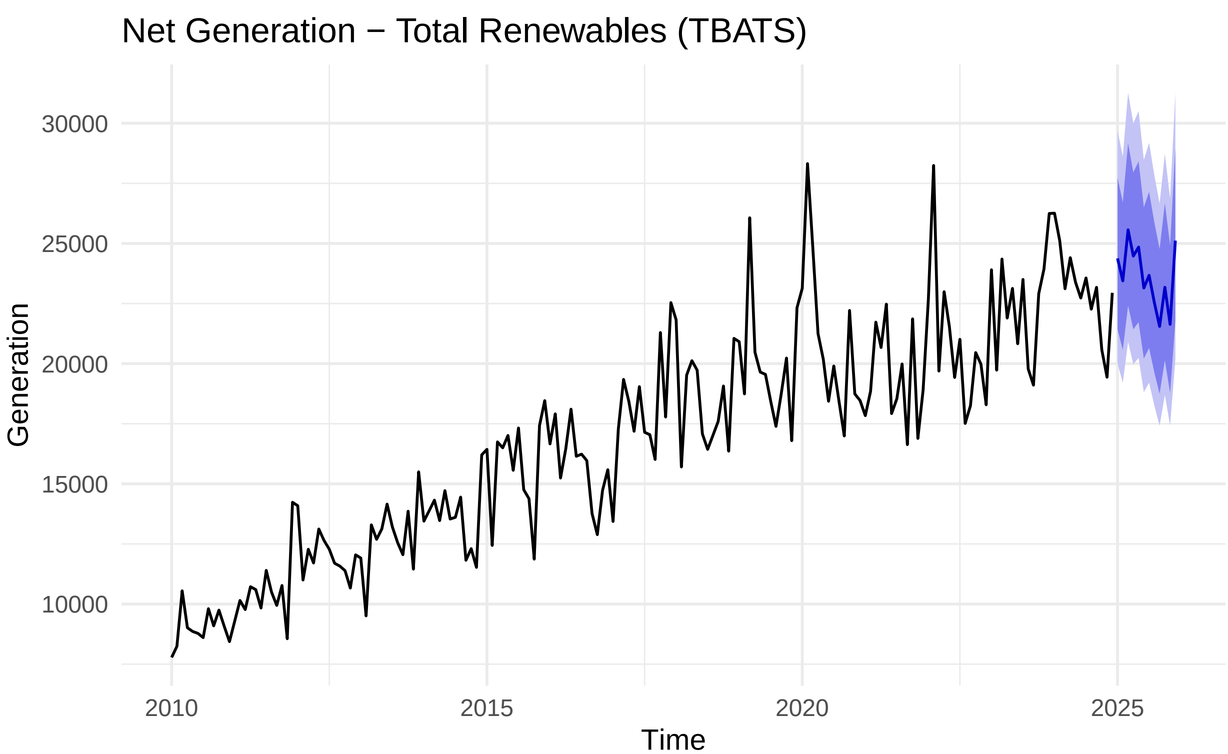
Plotting everything together





Summary and Conclusions

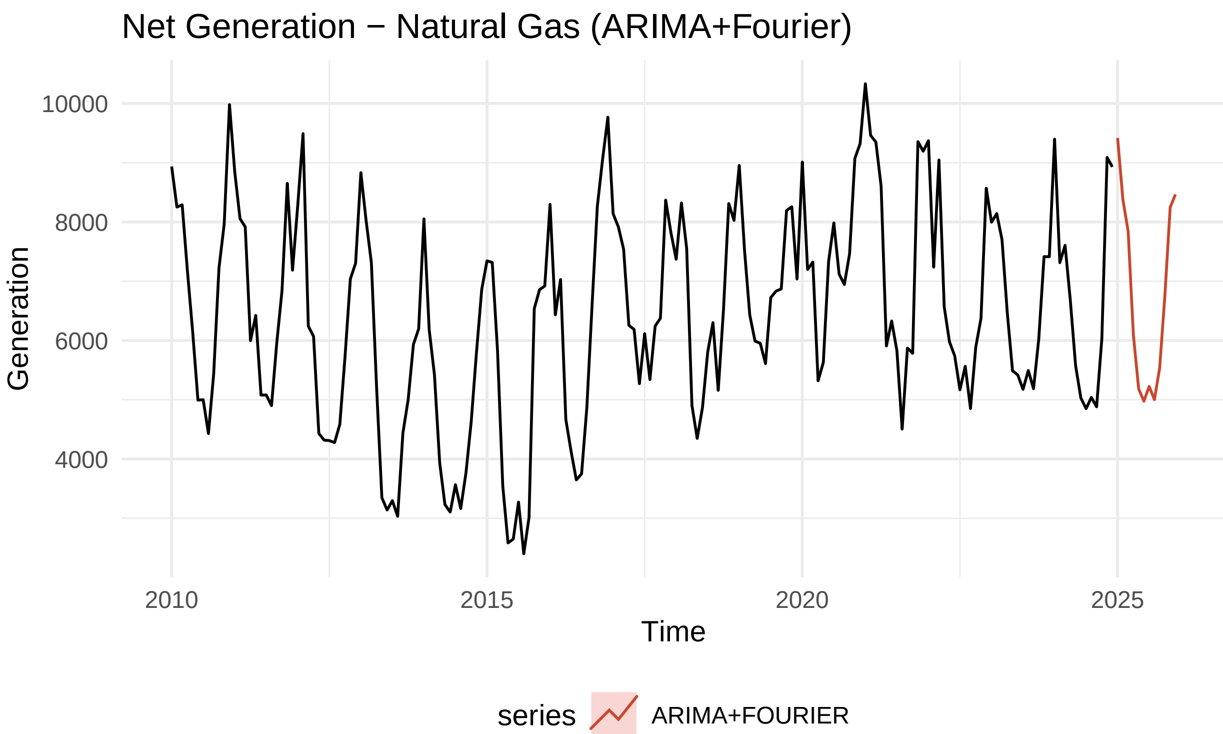
Final Forecast for Renewables: TBATS is the best-fit model



Based on testing with four advanced models—STL + ETS, ARIMA with Fourier terms, TBATS, and a Neural Network—it was found that the TBATS model provided the most robust fit for the renewables dataset. While STL + ETS and ARIMA+Fourier served as reasonable baselines, they underperformed in capturing the complex growth patterns and variable fluctuations in renewable generation. The Neural Network model also showed promise but suffered from slightly higher RMSE and residual autocorrelation, suggesting potential overfitting. In contrast, the TBATS model achieved the lowest RMSE (1933.02) and lowest residual autocorrelation (ACF1 = 0.093), indicating it captured both the long-term trend and the irregular seasonal dynamics more effectively than other methods.

Although all models slightly overpredicted renewable output, the forecasted values from TBATS showed a smooth continuation of the growth trend into 2025, consistent with the ongoing expansion of installed renewable capacity in Germany. Unlike natural gas, which showed strong cyclical behavior, the renewable generation data exhibited a dominant trend and relatively weak seasonality. The TBATS forecast reflects this by projecting continued, though slightly tapered, growth in renewable output.

Final Forecast for Natural Gas : ARIMA + Fourier is the best-fit model



For Natural Gas, it was found that the ARIMA model with Fourier terms provided the best fit. While the STL + ETS model serves as a good baseline, it falls short in capturing more nuanced patterns, such as those present in natural gas generation. Due to the strength of the ARIMA model with Fourier terms in handling strong seasonality, this study assumed it would be the best-performing model. The results confirmed this assumption, as it achieved the lowest RMSE and MAE values among the models tested.

Although the TBATS model fit the training data well, its forecasts were erratic, suggesting possible overfitting. TBATS and STL + ETS models tended to overpredict more than the other models, as indicated by their higher ME values. In fact, all models slightly overpredicted on average, given that all ME values were positive. The Neural Network (NN) model appeared to be too complex for the natural gas dataset, with higher RMSE and MAE values implying overfitting and poor generalization.

The forecasted values for 2025 showed little change before and after the onset of the Russia-Ukraine war. The natural gas dataset exhibited strong, consistent seasonality throughout the years, seemingly following normal seasonal trends. Although this study initially assumed that the war would have a more significant impact, the actual effect on natural gas usage was less than expected.

Takeaway

In conclusion, this study suggests that renewable energy will continue to play an expanding role in Germany's electricity grid, while natural gas is likely to maintain a stable role. The Russia-Ukraine war did not appear to significantly affect the energy mix in terms of natural gas usage for electricity generation but may impose possible accelerating effects on renewables deployment. However, the renewables generation growth rate is expected to slow down in 2025, which may be effected by other variables that are not examined in this study.

Limitations and Further Analysis

One limitation of this study is that it focused solely on the electricity mix component of natural gas usage, excluding its role in heating, which accounts for approximately 44% of Germany's imported natural gas consumption. This omission likely contributed to the observed minor changes in natural gas usage, as the impact on the electricity sector alone may not fully capture the broader implications of shifts in energy policy or geopolitical events.

For future research, it is recommended to conduct a comprehensive analysis of the heating sector, where more significant changes in natural gas usage are anticipated. Additionally, future studies should expand the scope to include an analysis of the overall electricity mix, incorporating other energy sources and relevant parameters to provide a more holistic understanding of the energy system.

Furthermore, it is suggested that future research explore forecasting not only the quantities of natural gas and renewable energy but also their price dynamics. This would provide deeper insights into market behavior and inform energy policy planning more effectively.

References

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